

Section A: Pure Mathematics [35 marks]

- 1 Find, algebraically, the set of values of k for which $(k-15)x^2 - 16x + k - 3 < 0$ for all real values of x . [4]

Solution

Two conditions must be satisfied:

$$(k-15) < 0 \text{ and } b^2 - 4ac < 0$$

[M1: coefficient]

$$(-16)^2 - 4(k-15)(k-3) < 0$$

[M1]

$$256 - 4(k^2 - 18k + 45) < 0$$

$$64 - k^2 + 18k - 45 < 0$$

$$k^2 - 18k - 19 > 0$$

$$(k+1)(k-19) > 0$$

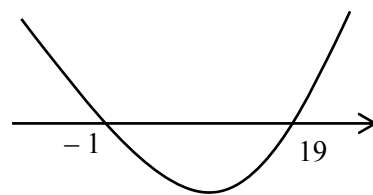
[M1]

Using GC, $k < -1$ or $k > 19$

Since $(k-15) < 0$, reject $k > 19$

Hence, the set of values is $\{k \in \mathbb{R} : k < -1\}$.

[A1]



- 2 Solve algebraically the simultaneous equations

$$\frac{1}{2}(2^y) = 4^{x+2},$$

$$\frac{3^y}{3^{xy}} = 27.$$

[4]

Solution

$$2^{y-1} = 2^{2(x+2)}$$

$$y-1 = 2x+4$$

[M1]

$$y = 2x+5 \quad \text{--- (1)}$$

$$3^{y-xy} = 3^3$$

$$y - xy = 3$$

[M1]

$$2x+5 - x(2x+5) = 3$$

[M1]

$$2x^2 + 3x - 2 = 0$$

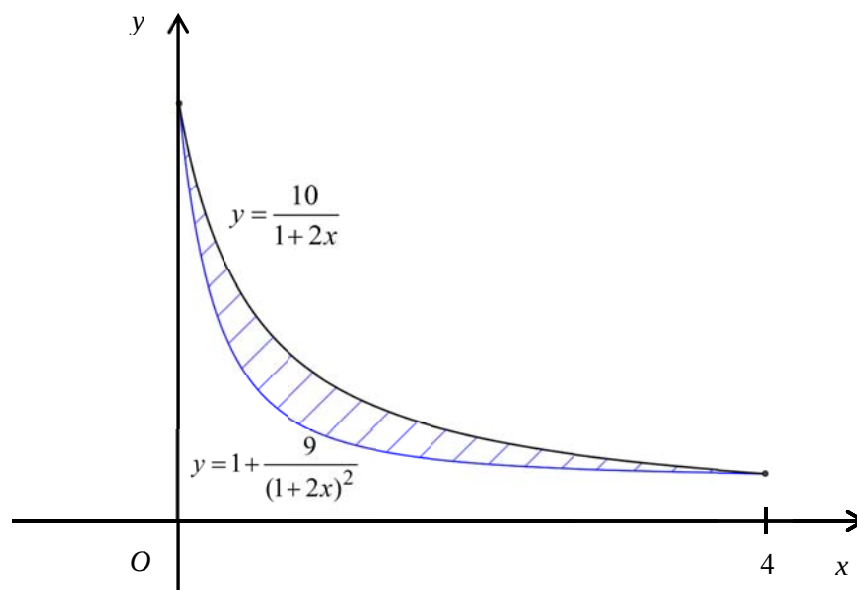
$$(2x-1)(x+2) = 0$$

$$\text{When } x = \frac{1}{2}, \quad y = 6$$

$$\text{When } x = -2, \quad y = 1$$

[A1]

3



The diagram shows a sketch of part of the curve $y = \frac{10}{1+2x}$ and part of the curve $y = 1 + \frac{9}{(1+2x)^2}$, with their points of intersection at $x = 0$ and $x = 4$.

Use integration to find the exact area of the shaded region.

[4]

Solution

Area

$$= \int_0^4 \left(\frac{10}{1+2x} - 1 - \frac{9}{(1+2x)^2} \right) dx$$

M1

$$= \left[5 \ln(1+2x) - x + \frac{9}{2(1+2x)} \right]_0^4$$

M1M1

$$= \left[5 \ln 9 - 4 + \frac{1}{2} \right] - \left[\frac{9}{2} \right]$$

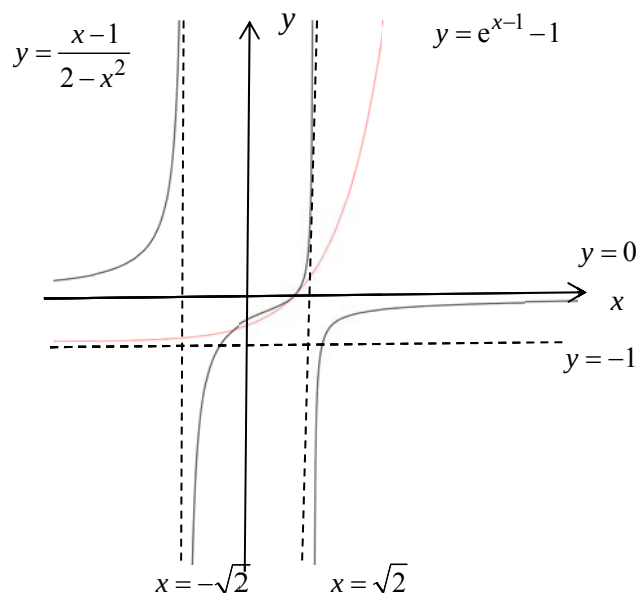
$$= 5 \ln 9 - 8 \text{ units}^2$$

A1

- 4 Sketch on a single diagram, the graphs of $y = \frac{x-1}{2-x^2}$ and $y = e^{x-1} - 1$, indicating clearly the equations of any asymptotes for each of the graphs. [3]

Hence find the range of values of x such that $\frac{x-1}{2-x^2} \geq e^{x-1} - 1$. [2]

Solution



Asymptotes at $x = -\sqrt{2}$, $x = \sqrt{2}$, $y = 0$ and $y = -1$.

G1G1B1

Intersection point at $x = -0.38790$.

The range of values: $x < -\sqrt{2}$ or $-0.38790 \leq x < \sqrt{2}$.

A1A1

- 5 (a) Find

(i) $\int e^{3-2x} dx$ [1]

(ii) $\int \left(\frac{x^2 - 2x}{\sqrt{x}} \right) dx$. [2]

(b) Show that $\frac{d}{dx} (e^{-x+\ln x}) = (1-x)e^{-x}$.

Hence show that $\int_1^4 (x-1)e^{-x} dx = \frac{1}{e^4} (e^3 - 4)$. [5]

Solution

(a)(i) $\int e^{3-2x} dx = -\frac{1}{2} e^{3-2x} + c$

B1

$$\begin{aligned}
 \text{(ii)} \quad & \int \left(\frac{x^2 - 2x}{\sqrt{x}} \right) dx \\
 &= \int \left(x^{\frac{3}{2}} - 2x^{\frac{1}{2}} \right) dx && \text{M1} \\
 &= \frac{2}{5} x^{\frac{5}{2}} - \frac{4}{3} x^{\frac{3}{2}} + c && \text{A1}
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad & \frac{d}{dx} \left(e^{-x+\ln x} \right) \\
 &= \left(e^{-x+\ln x} \right) \left(-1 + \frac{1}{x} \right) && \text{M1} \\
 &= \left(e^{-x} \cdot e^{\ln x} \right) \left(\frac{1-x}{x} \right) && \text{A1} \\
 &= (1-x) e^{-x}
 \end{aligned}$$

$$\begin{aligned}
 & \int_1^4 (x-1) e^{-x} dx \\
 &= - \left[e^{-x+\ln x} \right]_1^4 && \text{M1} \\
 &= -e^{-4+\ln 4} + e^{-1} && \text{M1} \\
 &= \frac{1}{e} - \left(\frac{1}{e^4} \cdot e^{\ln 4} \right) && \text{A1} \\
 &= \frac{1}{e^4} (e^3 - 4)
 \end{aligned}$$

- 6** The equation of a curve is $y = \ln\left(\frac{20x}{10-x}\right)$, where $0 < x < 10$.
- (i)** Show that $\frac{dy}{dx} = \frac{10}{x(10-x)}$. [2]
- (ii)** The gradient of the curve at the point P is $\frac{2}{5}$. Show that the y – coordinate of P is $\ln 20$ and hence find the exact equation of the tangent to the curve at P . [4]
- (iii)** The tangent to the curve at P cuts the x – axis at Q . Find the exact coordinates of Q . [1]
- (iv)** R is the point on the x -axis vertically below P . Show that the area of triangle PQR is $\frac{5}{4}(\ln 20)^2$. [3]

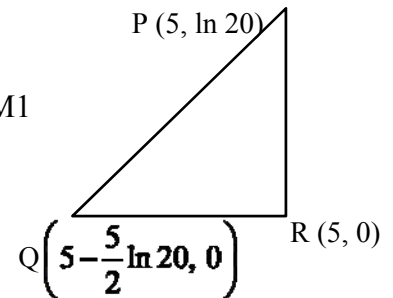
Solution

- (i)** $y = \ln\left(\frac{20x}{10-x}\right)$
 $y = \ln 20x - \ln(10-x)$ M1
 $\frac{dy}{dx} = \frac{1}{x} + \frac{1}{10-x}$ A1
 $\frac{dy}{dx} = \frac{10}{x(10-x)}$ (shown)
- (ii)** $\frac{10}{x(10-x)} = \frac{2}{5}$ M1
 $50 = 20x - 2x^2$
 $2x^2 - 20x + 50 = 0$
 $x^2 - 10x + 25 = 0$
 $(x-5)^2 = 0$ A1
 $x = 5$
 Sub. $x = 5$ into $y = \ln\left(\frac{20x}{10-x}\right)$, B1
 $y = \ln 20$ (shown)
 At $P(5, \ln 20)$, the equation of the tangent is
 $y - \ln 20 = \frac{2}{5}(x - 5)$ B1
 $y = \frac{2}{5}x - 2 + \ln 20$
- (iii)** At $Q, y = 0, x = 5 - \frac{5}{2}\ln 20$. $Q \equiv \left(5 - \frac{5}{2}\ln 20, 0\right)$. B1

(iv) $R(5, 0)$

M1

$$\begin{aligned} \text{Area of triangle } PQR &= \frac{1}{2} \left(5 - \left(5 - \frac{5}{2} \ln 20 \right) \right) (\ln 20) \quad \text{M1M1} \\ &= \frac{5}{4} (\ln 20)^2. \end{aligned}$$

**Section B: Statistics [60 marks]**

- 7 (a) A bag contains 3 red balls, 4 green balls and 5 blue balls. Three balls are drawn from the bag at random without replacement. Calculate the probability that

(i) all three balls are red; [1]

(ii) the third ball is red; [2]

(iii) the first two balls are of the same color given that the third ball is red. [3]

Are the events given in part (i) and part (ii) mutually exclusive? Explain your answer. [1]

- (b) The independent events A and B are such $P(A) = 0.6$ and $P(A' \cap B') = 0.3$. Show that $P(B)$ is $\frac{1}{4}$. [3]

Solution

There are 12 balls altogether, 3 are red and 9 are not red.

(i) $P(\text{all three balls are red})$

$$\begin{aligned} &= \frac{3}{12} \times \frac{2}{11} \times \frac{1}{10} \\ &= \frac{1}{220} \text{ or } 0.00455 \text{ (3 sf)} \quad \text{[A1]} \end{aligned}$$

(ii) $P(\text{the third ball is red})$

$$\begin{aligned} &= P(RRR) + P(RR'R) + P(R'RR) + P(R'R'R) \\ &= \frac{1}{220} + \frac{3}{12} \times \frac{9}{11} \times \frac{2}{10} + \frac{9}{12} \times \frac{3}{11} \times \frac{2}{10} + \frac{9}{12} \times \frac{8}{11} \times \frac{3}{10} \quad \text{[M1]} \\ &= \frac{1}{4} \text{ or } 0.25 \quad \text{[A1]} \end{aligned}$$

(iii) Let X and Y denote the following events:

X : the first two balls are the same color

Y : the third ball is red

$$P(X \cap Y) = P(RRR) + P(GGR) + P(BBR)$$

$$= \frac{1}{220} + \frac{4}{12} \times \frac{3}{11} \times \frac{3}{10} + \frac{5}{12} \times \frac{4}{11} \times \frac{3}{10} \quad [\text{M1}]$$

$$= \frac{17}{220} \text{ or } 0.0773 \text{ (3 sf)}$$

$$\therefore P(X|Y) = \frac{P(X \cap Y)}{P(Y)}$$

$$= \frac{\frac{17}{220}}{\frac{1}{4}} \quad [\text{M1}]$$

$$= \frac{17}{55} \text{ or } 0.309 \text{ (3 sf)} \quad [\text{A1}]$$

The 2 events are not mutually exclusive.

Let W denote the event “all three balls are the red”.

When W occurs, the third ball is red. Hence, $W \cap Y \neq \emptyset$.

$\therefore W$ and Y cannot be mutually exclusive. **[B1 – with correct reason]**

- (b) Since A and B are independent, $P(A \cap B) = P(A) \times P(B) = 0.6 P(B)$ M1
 $P(A \cup B) = 1 - P(A' \cap B') = 0.7$ M1
 $0.7 = P(A) + P(B) - P(A \cap B) = 0.6 + P(B) - 0.6 P(B)$
 $0.1 = 0.4 P(B)$ A1
Hence, $P(B) = \frac{1}{4}$ (shown)

- 8 A promotion at a restaurant entitles diners to a lucky draw for every meal purchased. Based on a survey of 500 randomly chosen diners last year, 411 did not win anything while the rest received a prize.

- (a) Nine students plan to have their dinner at the restaurant every weekday for one week this year.
- (i) Show that the probability that a randomly chosen student will first win a prize on his third meal at the restaurant is 0.120. State an assumption that you have used in your calculation. [2]
- (ii) Find the probability that less than three of the students will each first win a prize on their third meal at the restaurant. [2]
- (b) Using a suitable approximation, find the probability that at least 100 but less than 190 of the first 1000 diners this year will win a prize. [4]
- (c) A disgruntled diner decides to conduct a survey of n diners this year. Find the minimum value of n if the probability that at most 90% of the diners did not win a prize exceeds 0.75. [3]

Solution

- (i) P(a diner will first win a prize on his third meal at the restaurant)
 $= (0.822)^2(1 - 0.822)$ [M1]
 $= 0.12027$
 $= 0.120$ (3 sf)

Assumption: The figures from the survey last year still holds true for this year./The sample is representative/The probability of winning each day is the same/The event of winning on one day is independent of winning on another day/etc. [A1]

- (ii) Let X denote the random variable representing the number of students who will win a prize only on their third meal at the restaurant out of 9.
 $\therefore X \sim B(9, 0.120)$ [B1]
 $P(X < 3) = P(X \leq 2) = 0.91674$
 $= 0.917$ (3 sf) [A1]

- (iii) Let Y denote the random variable representing the number of diners, out of 1000, who will win a prize.
 $\therefore Y \sim B(1000, 0.178)$ [M1]
Since $n = 1000$ is large, $np = 178 > 5$, $nq = 822 > 5$,
 $Y \sim N(178, 146.316)$ approximately [M1]
 $P(100 \leq Y < 190) \xrightarrow{c.c.} P(99.5 \leq Y < 189.5)$ [M1]
 $= 0.82913$
 $= 0.829$ (3 sf) [A1]

- (iv) Let W denote the random variable representing the number of customers, out of n , that did not win a prize.
 $\therefore W \sim B(n, 0.822)$ [B1]
 $P(W \leq 0.9n) > 0.75$

Using the GC, we have

n	$P(W \leq 0.9n)$
7	0.746
8	0.792
9	0.829

$\therefore$ least $n = 8$

[M1]

[A1]

- 9** A recent survey by a student researcher, of 60 local companies, showed that four in five companies gave retired workers they re-employ the same pay as before. The records at the Labour Ministry, which tracks the practices of all companies in the country, revealed the actual figure to be 60% of all local companies instead.

- (i)** Give a reason for the difference in the student researcher's result if he had used systematic sampling in his selection of companies for the survey. [1]

A large company with 180 re-employed workers, agrees to help the student researcher in his research. He selected 60 re-employed workers from the company for a survey on the work that they do after re-employment. The satisfaction index, x , of each of the worker was recorded.

It was found that $\sum (x - 5) = 45.1$ and $\sum (x - 5)^2 = 1516$.

- (ii)** Explain how the student researcher could have selected the workers for the survey using the systematic sampling method. [2]
- (iii)** Find the probability that the mean satisfaction index of a sample of 60 randomly chosen workers lies between 5 and 7.5. [4]
- (iv)** State two assumptions or approximations used in your calculations in part **(iii)**. [2]

Solution

- (i) Even though the process of selection involves randomly choosing a number to select regular or evenly spaced out numbers in a list, for example every 10th company, the student researcher may have used a sampling frame that listed the companies in a particular manner that led to the over representation of companies that re-employ employees with the same pay.

[B1 - Correct explanation using a sampling frame with a “particular pattern”]

- (ii) Assign numbers to the workers students in a random manner from 1 to 60.

Since $\frac{180}{60} = 3$, randomly choose a number from 1 to 3, say r .

A systematic sample can be obtained by selecting the $r^{\text{th}}, (r + 3)^{\text{th}}, (r + 6)^{\text{th}}, \dots, (r + 57)^{\text{th}}$ worker.

[B2 - Correct explanations]

- (iii)

$$\bar{x} = \frac{45.1}{60} + 5$$

$$= 5.75167$$

$$= 5.75 \text{ (3 sf)}$$

[B1]

$$s^2 = \frac{1}{59} \left[1516 - \frac{(45.1)^2}{60} \right]$$

$$= 25.12034$$

$$= 25.1 \text{ (3 sf)}$$

[B1]

Since $n = 60$ is large, by CLT, $\bar{X} \sim N\left(5.75167, \frac{25.12034}{60}\right)$ approximately. **[B1]**

$$\therefore P(5 < \bar{X} < 7.5) = 0.87387$$

$$= 0.874 \text{ (3 sf)}$$

[A1]

- (iv) 1. Since n is large, Central Limit Theorem is used to approximate the distribution of the sample mean to be normal.
2. Assume that the satisfaction index of two randomly chosen workers will be independent.
3. Assume that the sample is representative of the population such that the unbiased estimates are good approximations to the population mean and variance.

[B2]

- 10** The mass, x kg, of the contents of each packet in a random sample of 60 cereal packets is measured, and the results are summarized by

$$\sum x = 59.6475, \sum (x - \bar{x})^2 = 15.9772.$$

- (i) Find the unbiased estimates of the population mean and variance. [2]
 (ii) Test, at the 5% significance level, whether the population mean mass of the contents is less than 1.10 kg. [3]
 (iii) Explain the meaning of “at the 5% significance level” in the context of the question. [1]

In another test, a new sample of 30 cereal packets yielded a sample mean of $\bar{x}$ kg. The results are to be used to test, at the 5% significance level, the hypothesis that the population mean mass of the contents differs from 1.10 kg. It may be assumed that the masses of the contents are normally distributed with standard deviation 0.53 kg. Find the range of values of $\bar{x}$ for which the null hypothesis would not be rejected. [4]

Solution

(i)

$$\bar{x} = \frac{\sum x}{n} = \frac{59.6475}{60} = 0.994125 \quad \text{M1}$$

$$\begin{aligned} s^2 &= \frac{1}{n-1} \left[\sum (x - \bar{x})^2 \right] \\ &= \frac{1}{59} [15.9772] \quad \text{M1} \\ &= 0.2708 \end{aligned}$$

(ii)

To test $H_0 : \mu = 1.10$

against $H_1 : \mu < 1.10$

Use one-tail Z-test at 5% significance level B1

Since $n = 60$ is large, by CLT

Under H_0 , $\bar{X} \sim N\left(1.10, \frac{0.2708}{60}\right)$ approximately B1

Using GC,

p -value = 0.0575 > 0.05, hence we do not reject H_0 and conclude that there is insufficient evidence that the population mean mass of the contents is less than 1.10 kg at 5% level of significance. A1

- (iii) The expression ‘at the 5% significance level’ refers to the probability of 0.05 of concluding that the mean mass of the contents is less than 1.10 kg when actually it was not. B1

Assume that the mass of the contents follows a normal distribution with standard deviation 0.53 kg.

To test $H_0 : \mu = 1.10$

against $H_1 : \mu \neq 1.10$

Use two-tail Z-test at 5% significance level

B1

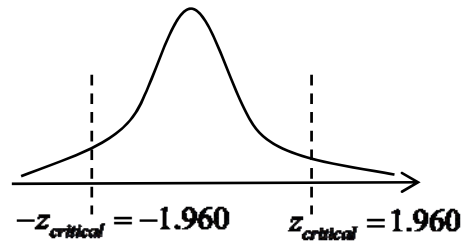
Under H_0 , $\bar{X} \sim N\left(1.10, \frac{0.53^2}{30}\right)$

Test statistic: $Z = \frac{\bar{X} - 1.10}{\frac{0.53}{\sqrt{30}}} \sim N(0, 1)$

B1

At 5% significance level, $z_{critical} = 1.960$

$$z_{test} = \frac{\bar{x} - 1.10}{\frac{0.53}{\sqrt{30}}}$$



For the null hypothesis not to be rejected,

$$-1.960 < \frac{\bar{x} - 1.10}{\frac{0.53}{\sqrt{30}}} < 1.960$$

M1

$$0.910 < \bar{x} < 1.29$$

A1

- 11** A supermarket sells apples and bunches of bananas. The weight of an apple is known to be normally distributed with mean 150 grams and standard deviation 10 grams. The weight of a bunch of bananas is known to be normally distributed with mean 780 grams and standard deviation 70 grams.

- (i) Find the probability that the total weight of ten randomly chosen apples will exceed twice the weight of a bunch of bananas. [3]

The supermarket sells the apples for \$0.46 each. Each bunch of bananas is sold by weight at a price of \$1.80 per kilogram.

- (ii) Find the probability that two randomly chosen bunches of bananas will cost more than six apples. [4]

- (iii) The supermarket now wants to grade its apples into Grade A, Grade B and Grade C by weight, with Grade A apples being the lightest and Grade C apples being the heaviest.

Find the range of the weight of a Grade B apple if the supermarket wants 15% of its apples to be of Grade A and 20% of its apples to be of Grade B. [3]

Solution

- (i) Let X denote the random variable representing the weight of an apple.
 $\therefore X \sim N(150, 10^2)$
 Let Y denote the random variable representing the weight of a bunch of bananas.

$$\therefore Y \sim N(780, 70^2)$$

$$\begin{aligned} E(X_1 + X_2 + \dots + X_{10} - 2Y) &= 10E(X) - 2E(Y) \\ &= 10(150) - 2(780) \\ &= -60 \end{aligned} \quad \text{[B1]}$$

$$\begin{aligned} \text{Var}(X_1 + X_2 + \dots + X_{10} - 2Y) &= 10\text{Var}(X) + 4E(Y) \\ &= 10(100) + 4(4900) \\ &= 20600 \end{aligned} \quad \text{[B1]}$$

$$\therefore X_1 + X_2 + \dots + X_{10} - 2Y \sim N(-60, 20600)$$

$$\begin{aligned} \text{From GC, } P(X_1 + X_2 + \dots + X_{10} - 2Y > 0) \\ &= 0.33796 \\ &= 0.338 \text{ (3 sf)} \end{aligned} \quad \text{[A1]}$$

- (ii) Let W denote the random variable representing the cost of two randomly chosen bunches of bananas.

$$\therefore W = \frac{1.8}{1000}(Y_1 + Y_2)$$

$$\begin{aligned} E(W) &= \frac{1.8}{1000}[E(Y) + E(Y)] = \frac{1.8}{1000} \times 2 \times 780 \\ &= 2.808 \end{aligned} \quad \text{[B1]}$$

$$\begin{aligned} \text{Var}(W) &= \frac{3.24}{1000000}[\text{Var}(Y) + \text{Var}(Y)] = \frac{3.24}{1000000} \times 2 \times 70^2 \\ &= 0.031752 \end{aligned} \quad \text{[B1]}$$

$$\therefore W \sim N(2.808, 0.031752)$$

$$\begin{aligned} \text{From GC, } P(W > 0.46 \times 6) & \quad \text{[M1]} \\ &= P(W > 2.76) \\ &= 0.60618 \\ &= 0.606 \text{ (3 sf)} \quad \text{[A1]} \end{aligned}$$

(iii) Let X denote the random variable representing the weight of an apple.

$$\therefore X \sim N(150, 10^2)$$

Suppose an apple is a Grade B apple if it weighs between a grams and b grams, then

$$P(X \leq a) = 0.15 \text{ and } P(X \leq b) = 0.35 \quad \text{[B2]}$$

$$\text{From GC, } a = 139.6357 = 140 \text{ (3 sf)}$$

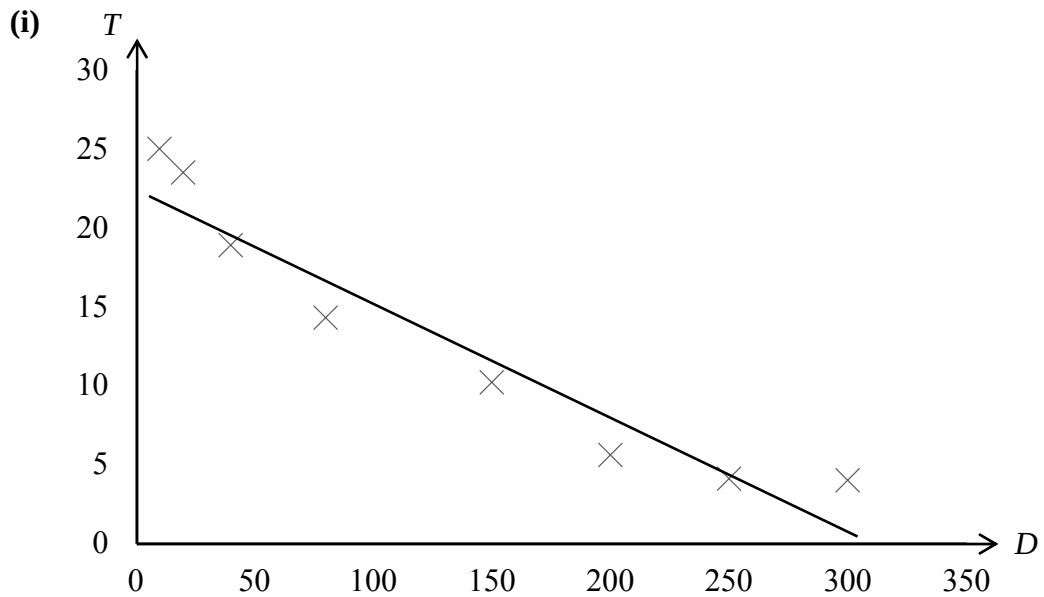
$$\text{and } b = 146.1468 = 146 \text{ (3 sf)}$$

$\therefore$ a Grade B apple weighs between 140 grams and 146 grams. [A1]

- 12** The water temperature T , in $^{\circ}\text{C}$, and the depth D , in metres, were recorded at noon on a certain day at each of 8 locations in a lake. The results are summarized in the table below.

D	10	20	40	80	150	200	250	300
T	25.0	23.5	18.9	14.3	10.2	5.6	4.1	4.0

- (i) Give a sketch of the scatter diagram for the data, as shown on your calculator. [2]
- (ii) Find the product moment correlation coefficient and comment on its value in the context of this question. [2]
- (iii) Find the equation of the regression line of T on D . Sketch this line on your scatter diagram. [2]
- (iv) Calculate an estimate of the water temperature at noon at a place in the region with depth 120 metres. Comment on the reliability of this estimate. [2]
- (v) It has been found that due to equipment error, all readings of temperature are 0.5°C higher than the actual water temperature. Without performing any further calculations, state with a reason, if you would expect a change in the value found in part (ii). [2]

Solution

[B1: points, B1: axes]

- (ii) By GC, $r = -0.960$ (3 s.f.) [B1]
 Since the r -value is close to -1 , there is a strong negative linear correlation between the water temperature and the depth. This suggests that the deeper in the lake it is, the lower the temperature of the water. [B1]
- (iii) Using GC, $T = 22.94945 - 0.07428155D = 22.9 - 0.0743D$ (3 s.f.) [B1, B1]
- (iv) $T = 22.94945 - 0.07428155(120) = 14.0$ (3 s.f.) [B1]
 The temperature is 14.0 degrees Celsius.
 The estimate is reliable as $D = 120$ is within the data range so interpolation is carried out. Furthermore, the r value is close to -1 which suggests a strong negative linear correlation within the data range. [B1]
- (v) The value would remain unchanged. [B1]
 As the actual values are all reduced by the same amount, the degree of scatter remains the same. Hence, the r value will remain unchanged. [B1]